RAMSES Data Structure and the LaRT Interface

Contents

1. Overview of RAMSES **1**

2. Output Directory Layout **2**

3. The Info File (`info_NNNNN.txt`) **2**

4. The AMR File (`amr_NNNNN.outCCCCC`) **3**

4.1 Header Records 3

4.2 Level Blocks 4

4.3 Oct Structure and Leaf Identification 5

4.4 Cell Position Calculation 5

5. The Hydro File (`hydro_NNNNN.outCCCCC`) **5**

5.1 Header Records 6

5.2 Variable Ordering 6

6. Physical Unit Conversions **7**

6.1 Hydrogen Number Density 7

6.2 Velocity 7

6.3 Temperature 7

7. The LaRT Reader: Step-by-Step **7**

7.1 Domain Skipping 8

8. File Naming Convention **8**

9. LaRT Input for RAMSES Data **8**

10. Generic Text Format vs. RAMSES Binary **9**

11. Known Limitations and Assumptions **10**

1. Overview of RAMSES

RAMSES [1] is an adaptive mesh refinement (AMR) code for astrophysical hydrodynamics and N-body dynamics. It decomposes the computational domain into an octree hierarchy of grids, refining regions of

interest to arbitrarily high resolution. The MPI decomposition uses a space-filling curve (Peano–Hilbert ordering) to distribute octs among CPU ranks.

`LaRT v2.00` can read RAMSES binary output directly via `amr_type = 'ramses'`. This document describes the RAMSES binary format in detail and explains how `read_ramses_amr.f90` translates it into the flat leaf-cell arrays required by the LaRT octree engine.

2. Output Directory Layout

A RAMSES snapshot is stored in a directory named `output_NNNNN/` where `NNNNN` is the five-digit output number. For a run with N_{cpu} MPI tasks the snapshot contains the following per-CPU files:

```
output_00010/
    info_00010.txt            ! global metadata: units, cosmology, etc.
    amr_00010.out00001        ! AMR tree for CPU 1
    amr_00010.out00002        !   ...
    ...
    amr_00010.outNNNNN        ! AMR tree for CPU N
    hydro_00010.out00001      ! hydrodynamic variables for CPU 1
    hydro_00010.out00002
    ...
    hydro_00010.outNNNNN
    gravity_00010.out00001    ! gravitational potential (optional)
    part_00010.out00001       ! particle data (optional)
```

All `.out*` files are Fortran *unformatted sequential* binary files (each record bracketed by 4-byte integer record markers on little-endian systems).

3. The Info File (`info_NNNNN.txt`)

`info_NNNNN.txt` is a plain-text file containing global simulation parameters. The fields relevant to LaRT are listed below.

Key	Type	Meaning
`ncpu`	integer	Number of MPI ranks used for this snapshot
`ndim`	integer	Number of spatial dimensions (always 3 for 3-D runs)
`nlevelmax`	integer	Maximum AMR refinement level used
`boxlen`	real	Box length in *code units* (see below)
`unit_l`	real	Length unit: 1 code length = `unit_l` cm
`unit_d`	real	Density unit: 1 code density = `unit_d` g cm^{-3}
`unit_t`	real	Time unit: 1 code time = `unit_t` s
`gamma`	real	Adiabatic index γ of the EOS

The velocity unit is derived as `unit_v` = `unit_l`/`unit_t` (cm s^{-1}).

Cosmological runs. For cosmological simulations, `boxlen` is in Mpc/h and `unit_l` incorporates the scale factor a so that $L_{\rm phys}$ = `boxlen` × `unit_l` cm is the comoving box size. LaRT reads the physical box size as $L_{\rm cm}$ = `boxlen` × `unit_l`.

Example info file (non-cosmological).

```
ncpu        =           64
ndim        =            3
nlevelmax   =           12
ngridmax    =       500000
boxlen      =   1.0000000000000000E+02
unit_l      =   3.0856775814913670E+21   ! 1 kpc in cm
unit_d      =   6.7707722919716770E-25
unit_t      =   3.0856775814913670E+14
gamma       =   1.6666666666666667E+00
```

In this example the box is 100 kpc on a side.

4. The AMR File (`amr_NNNNN.outCCCCC`)

Each CPU file stores the AMR tree data for that CPU's domain. The file consists of a *header section* followed by a sequence of *level blocks*, one per AMR level.

4.1 Header Records

The AMR header records are read in the following order:

Record	Variable(s)	Description
1	`ncpu`	Total number of MPI ranks
2	`ndim`	Number of dimensions
3	`nx, ny, nz`	Coarse grid size (typically $1 \times 1 \times 1$ or $2 \times 2 \times 2$)
4	`nlevelmax`	Maximum AMR level
5	`ngridmax`	Maximum number of grids (octs) per CPU
6	`nboundary`	Number of boundary domains
7	`ngrid_current`	Current number of active grids
8	`boxlen`	Box length (code units)
9–21	(various)	Cosmological parameters, time, etc. (skipped by LaRT)
22	`ngridlevel(:,:)`	Grid count array: `(ncpu, nlevelmax)`
23	(level sums)	Skipped
24–27	(boundary grids)	Only present if `nboundary > 0` (skipped)
28–32	(ordering arrays)	Peano–Hilbert keys, son maps, etc. (skipped)

The ngridlevel array. `ngridlevel(j, l)` gives the number of octs belonging to CPU j at AMR level l. LaRT constructs a combined table `ngridfile(j, l)` that includes both domain and boundary grids.

4.2 Level Blocks

After the header, the file contains one block per AMR level $l = 1, \ldots, \texttt{nlevelmax}$. Within each level, data are stored *domain by domain*: first all domains $j = 1, \ldots, \texttt{ncpu}$, then all boundary domains $j = \texttt{ncpu} + 1, \ldots, \texttt{ncpu} + \texttt{nboundary}$.

For each domain j at level l (if $\texttt{ngridfile}(j, l) > 0$):

Record(s)	Variable	Description
1	`ind_grid(:)`	Global grid indices
2	`next(:)`	Linked-list next pointer
3	`prev(:)`	Linked-list previous pointer
4–6	`xg(:, 1:3)`	Oct centre coordinates in code units
7	`father(:)`	Parent grid index
8–13	`nbor(:,1:6)`	Neighbour grid indices (6 faces)
14–21	`son(:,1:8)`	Child grid indices; = 0 if a child is a **leaf**
22–29	`cpu_map(:,1:8)`	CPU ownership per child
30–37	`ref_map(:,1:8)`	Refinement flags per child

Record counts assume `ndim`= 3, `twotondim`= 8.

4.3 Oct Structure and Leaf Identification

RAMSES groups cells into *octs*: cubes of $2^3 = 8$ cells sharing a common parent grid cell. Each oct is identified by its *grid index*. Within an oct, the eight child cells are indexed $\texttt{ind} = 1, \ldots, 8$ with the following ordering:

$$i_x = (\texttt{ind} - 1) \bmod 2, \quad i_y = \lfloor (\texttt{ind} - 1)/2 \rfloor \bmod 2, \quad i_z = \lfloor (\texttt{ind} - 1)/4 \rfloor, \tag{1}$$

so that `ind`=1 is the $(-x, -y, -z)$ corner and `ind`=8 is the $(+x, +y, +z)$ corner.
A child cell at index `ind` in grid g at level l is a **leaf** if and only if

$$\texttt{son}(g, \texttt{ind}) = 0. \tag{2}$$

Otherwise the value of `son` is the grid index of the finer oct that occupies that child cell.

4.4 Cell Position Calculation

The centre of the coarse grid of the root domain is the origin. For a non-cosmological run with `nx=ny=nz=1`:

$$\mathbf{x}_{\text{bound}} = \left(\frac{n_x}{2}, \frac{n_y}{2}, \frac{n_z}{2} \right). \tag{3}$$

The cell size at level l is $\Delta x_l = 2^{-l}$ (in units where $\texttt{boxlen} = 1$). The offset of child cell `ind` from the oct centre is

$$\boldsymbol{\delta}_{\texttt{ind}} = \left(i_x - \tfrac{1}{2},\ i_y - \tfrac{1}{2},\ i_z - \tfrac{1}{2} \right) \times \Delta x_l. \tag{4}$$

The absolute position of child `ind` in grid g in fractional box units $[0, 1]$ is

$$\tilde{x} = \frac{x_g + \delta_{\texttt{ind},x} - x_{\text{bound},x}}{n_x} + \frac{1}{2}, \tag{5}$$

and similarly for $\tilde{y}$ and $\tilde{z}$ (the $+1/2$ shifts the origin from the grid corner to the box corner). In the code:

```
xleaf = (xg(j,1) + xc(ind,1) - xbound(1)) / dble(nx) + 0.5d0
```

After reading, leaf positions are converted to physical units (cm):

$$x_{\text{phys}} = \tilde{x} \times L_{\text{box}}, \quad L_{\text{box}} = \texttt{boxlen} \times \texttt{unit_l}. \tag{6}$$

5. The Hydro File (`hydro_NNNNN.outCCCCC`)

The hydro file stores the conserved hydrodynamic variables for every cell (leaf and internal alike). It is organised in strict lockstep with the AMR file: one block per level per domain.

5.1 Header Records

Record	Variable	Description
1	`ncpu`	Number of MPI ranks
2	`nvar`	Number of hydro variables per cell
3	`ndim`	Number of dimensions
4	`nlevelmax`	Maximum AMR level
5	`nboundary`	Number of boundary domains
6	`gamma`	Adiabatic index

5.2 Variable Ordering

Standard RAMSES hydro stores the following variables in order:

Index	Variable	Physical meaning
1	ρ	Gas mass density [code units]
2	ρv_x	x-momentum density [code units]
3	ρv_y	y-momentum density [code units]
4	ρv_z	z-momentum density [code units]
5	E	Total energy density (kinetic + thermal) [code units]
6	$Z/Z_\odot$	Metallicity (if `metal=.true.`)
7+	passive scalars	Additional tracers (simulation-dependent)

Level block structure. For each level l and domain j, the hydro file contains:

1. Two header records (level number and domain number) — skipped.
2. For each cell index `ind`$= 1, \ldots, 8$ and each variable `ivar`$= 1, \ldots,$ `nvar`: one record of length `ngridfile`(j, l) containing the values for all grids of domain j at level l.

All records have the same element count `ngridfile`(j, l), so the loop structure is:

```
do ind = 1, twotondim      ! 8 cells per oct
  do ivar = 1, nvar        ! variables
    read(iu_hyd) var(:, ind, ivar)  ! (ngrida) values
  end do
end do
```

Precision. RAMSES writes hydro variables in double precision (`real*8`) by default. Some configurations compile with single precision (`real*4`). The LaRT reader handles both via the flag `ramses_hydro_prec` (set to 8 or 4 in `read_ramses_amr.f90`).

6. Physical Unit Conversions

6.1 Hydrogen Number Density

The mass density in code units is converted to CGS and then to hydrogen number density assuming a fully ionised gas with mean molecular weight $\mu_{\rm H} \approx 1$ (one proton mass per hydrogen nucleus):

$$n_H\ [\mathrm{cm}^{-3}] = \frac{\rho_{\rm code} \times \texttt{unit_d}}{m_H}, \quad m_H = 1.6726 \times 10^{-24}\ \mathrm{g}. \tag{7}$$

The neutral fraction $n_{\rm HI}/n_H$ is computed separately in `grid_mod_amr.f90` using collisional ionisation equilibrium (CIE) if `par%use_cie_condition = .true.`, otherwise all hydrogen is assumed neutral.

6.2 Velocity

RAMSES stores momentum density ρv_α (variable indices 2–4). The velocity in CGS is:

$$v_\alpha\ [\mathrm{cm\,s^{-1}}] = \frac{(\rho v_\alpha)_{\rm code}}{\rho_{\rm code}} \times \texttt{unit_v}, \quad \texttt{unit_v} = \frac{\texttt{unit_l}}{\texttt{unit_t}}. \tag{8}$$

LaRT receives velocities in km s^{-1} (divided by 10^5).

6.3 Temperature

RAMSES variable 5 is the *total energy density* E (kinetic plus thermal). The thermal energy density is:

$$e_{\rm th} = E - \frac{1}{2}\rho\, v^2 = E - \frac{(\rho v_x)^2 + (\rho v_y)^2 + (\rho v_z)^2}{2\rho}. \tag{9}$$

The specific thermal energy (per unit mass) in physical units is:

$$\varepsilon\ [\mathrm{erg\,g^{-1}}] = \frac{e_{\rm th}}{\rho_{\rm code}} \times \texttt{unit_v}^2. \tag{10}$$

The temperature follows from the ideal gas law with adiabatic index γ and mean molecular weight $\bar{\mu}$ (LaRT assumes $\bar{\mu} = 1.22$ for neutral gas):

$$T\ [\mathrm{K}] = (\gamma - 1)\, \varepsilon\, \frac{\bar{\mu}\, m_H}{k_B}, \quad k_B = 1.381 \times 10^{-16}\ \mathrm{erg\,K^{-1}}. \tag{11}$$

A floor of $T = 10$ K is applied after conversion.

7. The LaRT Reader: Step-by-Step

The reader in `read_ramses_amr.f90` proceeds as follows:

1. **Read `info` file.** Extract `ncpu`, `unit_l`, `unit_d`, `unit_t`, `boxlen`, and γ.

2. **Count leaf cells (first pass).** `ramses_count_leaves` opens every `amr_*.outCCCCC` file, reads the `son` arrays for all levels, and counts cells where `son` $= 0$. This gives the total $N_{\rm leaf}$ needed to pre-allocate output arrays.

3. **Read leaf data (second pass).** `ramses_read_all_cpus` opens *both* the AMR and hydro files simultaneously for each CPU:
   - Reads AMR grids level by level, extracting oct centres `xg` and the `son` array.
   - Reads hydro variables for the same grids.
   - For each child cell with `son` = 0 (leaf): computes its physical position, n_H, T, and $\mathbf{v}$, and appends them to the output arrays.
4. **Convert positions.** After the second pass, leaf positions (stored as fractions of `boxlen`) are multiplied by $L_{\rm cm}$ = `boxlen` × `unit_l` to give physical positions in cm. These are then passed to `amr_build_tree` as-is (with `distance2cm = 1`).
5. **Build octree and normalise.** Handled by `grid_mod_amr.f90` (see the AMR description document).

7.1 Domain Skipping

The AMR and hydro files interleave data from all $N_{\rm cpu}$ domains and boundary domains at each level. The LaRT reader only *stores* data for the domain that matches `icpu` (the loop variable); data from other domains are skipped with `skip_records`. This reproduces the original RAMSES read pattern used in post-processing tools such as PYMSES and YT.

8. File Naming Convention

File pattern	Content
`output_NNNNN/info_NNNNN.txt`	Global metadata
`output_NNNNN/amr_NNNNN.outCCCCC`	AMR tree for CPU `CCCCC`
`output_NNNNN/hydro_NNNNN.outCCCCC`	Hydro variables for CPU `CCCCC`
`output_NNNNN/gravity_NNNNN.outCCCCC`	Gravitational potential (optional)
`output_NNNNN/part_NNNNN.outCCCCC`	Particle data (optional)

where `NNNNN` is the zero-padded 5-digit snapshot index and `CCCCC` is the zero-padded 5-digit CPU rank (1-based).

9. LaRT Input for RAMSES Data

```
&parameters
 par%use_amr_grid  = .true.
 par%amr_type      = 'ramses'
```

```
 par%amr_file      = '/path/to/simulation'  ! repository directory
 par%amr_snapnum   = 10                     ! snapshot number (NNNNN)
 par%no_photons    = 1e6
 par%taumax        = 1.0e7
 par%DGR           = 0.0
 par%spectral_type = 'voigt'
 par%source_geometry = 'point'
 par%xs_point      = <x_cm>   ! source position in cm
 par%ys_point      = <y_cm>
 par%zs_point      = <z_cm>
 par%nxfreq  = 121
 par%nxim    = 64
 par%nyim    = 64
/
```

Notes.

- `par%amr_file` is the *repository directory* containing `output_NNNNN/` (not the output directory itself).
- `par%amr_snapnum` selects the snapshot (the integer `NNNNN`).
- For RAMSES data, `par%distance2cm` is automatically set to 1.0 because the reader returns positions in cm (pre-multiplied by `unit_l`). The `par%distance_unit` parameter is ignored.
- `par%taumax` is specified as the line-centre optical depth from the box centre to the $+z$ boundary.
- Source position (`xs_point` etc.) must be given in cm, matching the units of the RAMSES leaf positions after conversion.
- If `par%use_cie_condition = .true.`, the neutral fraction $n_{\rm HI}/n_H$ is computed from the temperature via collisional ionisation equilibrium; otherwise all hydrogen is assumed neutral.

10. Generic Text Format vs. RAMSES Binary

For cases where RAMSES binary output is unavailable (e.g., data from other simulation codes, or hand-constructed test grids), LaRT provides the generic text format described separately. The two formats differ in the following ways:

Property	RAMSES binary	Generic text
File format	Fortran unformatted binary	Plain text
Position unit	cm (auto, from `unit_l`)	Code units (set by `distance_unit`)
`par%amr_type`	`'ramses'`	`'generic'`
`par%amr_snapnum`	required	not used
Multi-CPU support	yes (reads all `.outCCCCC` files)	no (single file)
Physical units	full CGS conversion built in	user's responsibility

The generic text file format is:

```
nleaf  boxlen
x  y  z  level  nH[cm^-3]  T[K]  vx[km/s]  vy[km/s]  vz[km/s]
...
```

The Python utility `examples/python/make_amr_grid.py` provides the `AMRGrid` class to construct and write these files from arbitrary density, temperature, and velocity fields.

11. Known Limitations and Assumptions

1. **Neutral fraction.** The RAMSES reader sets n_H (total hydrogen density, not $n_{\rm HI}$). The neutral fraction is computed later in `grid_create_amr` using either CIE (if `par%use_cie_condition = .true.`) or the assumption of full neutrality. For simulations with radiation feedback or on-the-fly ionisation, a custom neutral fraction field from a passive scalar may be needed.

2. **Mean molecular weight.** The temperature calculation in `ramses_read_all_cpus` assumes $\bar{\mu} = 1.22$ (fully neutral primordial gas). For ionised or metal-enriched gas this gives an overestimate.

3. **Hydro precision.** The variable `ramses_hydro_prec` (default 8) must be changed to 4 for simulations compiled with single-precision hydro.

4. **Non-standard variable ordering.** If the RAMSES simulation includes passive scalars before the energy variable (e.g., some MHD setups), the variable index for energy (E = var 5) may be wrong. Check the RAMSES namelist (`hydro.nml`) for the actual ordering.

5. **Memory.** All leaf cells are broadcast to every MPI rank in LaRT. For large RAMSES outputs ($\gtrsim 10^7$ leaf cells) this may require significant RAM per node.

References

[1] Teyssier, R. 2002, A&A, 385, 337 (*RAMSES: a new high-resolution n-body and hydrodynamics code for cosmological simulations*)